Aegis-v2: Detection-Gated, Layer-Selective Self-Repair of Onboard Earth-Observation Classifiers under Orbit-Calibrated Radiation Upsets

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***Abstract*—Deep neural networks now run on commercial off-the-shelf (COTS) processors aboard Earth-observation satellites, where single-event upsets (SEUs) from trapped protons silently corrupt the stored weights. In IEEE-754 single precision, one flip of the most significant exponent bit turns a weight of order one into a value near 10^{38}, collapsing a classifier to chance. Existing remedies sit at two extremes: blind periodic scrubbing reloads the entire model on a timer, and in-situ software mitigation (activation clipping, batch-norm re-centering) leaves the corrupted weights in place. We present Aegis-v2, a detection-gated, layer-selective repair loop. A label-free behavioral monitor (an activation-range envelope and a batch-norm drift test) proposes suspect blocks on each incoming image batch; a CRC-32 comparison against golden checksums confirms which blocks actually changed, including the block just upstream of the earliest alarm; only confirmed blocks are restored, bit-exactly, from a compressed golden copy in radiation-tolerant storage. An optional periodic CRC audit catches faults too small to change behavior. We evaluate Aegis-v2 in a paired, seeded mission simulator that flies a real Sentinel-2A track (SGP4), defines the South Atlantic Anomaly physically from IGRF field strength, and calibrates upset rates to eight years of Alsat-1 flight data (4.04×10^{-7} upsets/bit/day). Over three simulated days and three radiation seeds on EuroSAT, Aegis-v2 keeps a ResNet-18 at 87.4% mean accuracy (clean 89.3%) versus 37.0% for 12-hour scrubbing, 38.5% for range clipping and 14.0% unprotected, winning every paired seed; matching its accuracy by scrubbing would take $4.7\times$ the repair traffic. A held-out-trained MobileNetV2 shows the same ordering at 3.4 MB/day. We also report a negative result: always-on monitoring makes Aegis-v2 cost 5–$22\times$ more energy than slow scrubbing under our energy model, so its advantage is bandwidth and availability, not energy. We document the evaluation pitfalls that invalidated our own early results and release the simulator, campaigns and an interactive dossier.**

***Index Terms*—Single-event upset, soft errors, fault tolerance, onboard machine learning, Earth observation, deep neural networks, self-repair, South Atlantic Anomaly, scrubbing, batch normalization.**

I. Introduction

ONBOARD inference has moved from demonstration to routine. Φ-Sat-1 flew a convolutional cloud screener on a COTS vision accelerator in 2020 and discarded cloudy frames before downlink [1]; OPS-SAT trained and replaced models in orbit [2]; and current small-satellite payloads carry Jetson-class GPUs and neural accelerators [3], [4]. The appeal is plain: the downlink is the bottleneck of Earth observation, so deciding onboard what is worth sending multiplies the useful data a satellite returns. The cost is reliability. COTS processors and memories are not radiation hardened, and in low Earth orbit (LEO) energetic protons trapped in the inner Van Allen belt, concentrated over the South Atlantic Anomaly (SAA) [5], deposit enough charge to flip stored bits [6], [7].

A neural network is an unusually fragile thing to store in such memory. Its knowledge lives in millions of floating-point weights, and the IEEE-754 binary32 format [8] packs an 8-bit exponent beside the sign. Trained weights are small, so their exponent's most significant bit is almost always zero; flipping it multiplies the weight by $2^{128} \approx 3.4 \times 10^{38}$. One such upset in a convolution kernel saturates every activation downstream and drives the classifier to chance, as targeted bit-flip attacks showed on purpose [9]. Upsets accumulate: unlike a transient glitch in logic, a flipped bit in SRAM or DRAM stays flipped until something rewrites it.

Spacecraft engineering answers with hardware: error-detecting and correcting codes (EDAC), triple modular redundancy and memory scrubbing, the periodic rewriting of memory from a trusted copy [7], [10]. Scrubbing is simple and robust but blind: it pays to rewrite the whole model whether or not anything is wrong, and between scrubs the model runs corrupted. The machine-learning community answers with software. Range restriction (Ranger [11], FT-ClipAct [12]) clamps activations to profiled bounds; BLINK [13] detects batch-norm drift and re-centers activations; RADAR [14] and MILR [15] detect or algebraically recover corrupted weights. These methods mitigate symptoms in place but, with the exception of algebraic recovery, leave the corrupted weights in memory, where further upsets compound them. A third option is to wait for a ground station and re-upload the model, which costs scarce uplink time and leaves the model broken until the next contact.

This paper asks whether a satellite can do better by combining the two worlds: use cheap, label-free signals from the network's own activity to notice damage, use exact checksums to prove which stored blocks changed, and then restore only those blocks from an onboard golden copy. We call this *detection-gated, layer-selective restore* and build it as Aegis-v2. The idea is simple; demonstrating it honestly is not. Evaluations of DNN resilience are easy to get wrong in ways that flatter the method: unrealistic fault models (zeroed weights instead of exponent flips) [16], synthetic inputs, monitors whose thresholds fire on every clean batch, energy budgets that omit

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Code, campaign results and the interactive mission dossier are available in the project repository (`Rg9906/aegis-v2`).

the monitor itself. Our own first prototype reported a 98.9% energy saving that did not survive correction (Section VIII). We therefore treat the evaluation methodology as a first-class contribution.

Contributions.

1) **A detection-gated, layer-selective repair loop** (Aegis-v2) that combines a two-tier label-free behavioral monitor with CRC-32 confirmation, including an *upstream-block rule* that localizes batch-norm parameter faults whose symptoms first appear one block downstream, and an optional periodic CRC audit for sub-threshold faults (Section V).
2) **A paired, orbit-calibrated mission simulator.** Faults are placed uniformly over stored bits at rates integrated minute by minute along a real Sentinel-2A track, with the SAA defined physically from IGRF field strength and the orbit-averaged rate calibrated to Alsat-1 flight data. Every strategy replays the identical fault stream, so comparisons are paired (Section IV).
3) **A head-to-head evaluation** against unprotected inference, range clipping, BN re-centering, periodic scrubbing at five intervals, and a ground-uplink reload, on two architectures with 95% intervals over seeds, measuring accuracy over time, repair traffic and modeled energy (Section VII).
4) **An honest cost accounting with a negative result**: once the monitor's compute is charged, Aegis-v2 is not an energy saver; it wins on bandwidth and availability (Section VII-D).
5) **A catalogue of evaluation pitfalls**, each of which silently invalidated a result during this work, with the invariant that prevents it (Section VIII).

II. Background

A. Radiation in Low Earth Orbit

In LEO the dominant source of SEUs in memory is the flux of trapped protons in the inner radiation belt, supplemented by galactic cosmic rays and solar particle events [7]. Because Earth's dipole is offset and tilted, the inner belt dips to its lowest altitude over the South Atlantic, where the field strength at a given altitude is weakest; this is the SAA [5]. A sun-synchronous satellite crosses it on several consecutive orbits each day, and most memory upsets on such missions are recorded inside it. The rate also varies with the solar cycle; on Alsat-1 the yearly mean rose from 41 to 119 upsets/day over eight years [10].

The Alsat-1 record is unusually useful for calibration: between November 2002 and August 2010 its commercial SRAM logged 265,649 upsets in 2,622 days, i.e. 4.04×10^{-7} upsets/bit/day, of which 98.55% were single-bit upsets (SBUs) and the remainder multiple-bit upsets (MBUs) [10].

B. Why Floating-Point Weights Are Fragile

A binary32 value is $v = (-1)^s \, 2^{e-127} \, (1 + f)$ with sign s, biased exponent $e \in [1, 254]$ for normal numbers, and fraction $f \in [0, 1)$ (Fig. 1). A flip in mantissa bit k changes v by a factor in $[1 - 2^{k-23}, \, 1 + 2^{k-23}]$, so most mantissa flips are invisible to a classifier. A flip in exponent bit $j \in [23, 30]$ multiplies v by $2^{\pm 2^{j-23}}$. For trained weights $|v| < 2$ means $e \leq 127$, so bit 30 is zero and a flip sets it, multiplying by exactly 2^{128}; if the result exceeds the format's range it becomes $\pm\infty$. A sign flip on a batch-norm running variance makes $\sqrt{\sigma^2 + \epsilon}$ undefined and yields NaN. Under uniform placement, about one hit in 32 lands on bit 30; we recorded 4.89 such hits per day on ResNet-18 against 4.6 expected.

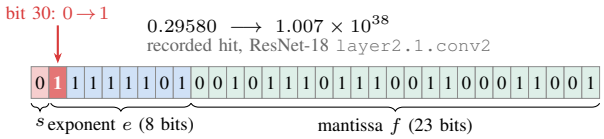


Fig. 1. IEEE-754 binary32 layout of a weight hit during our campaign (ResNet-18, seed 1). Flipping the most significant exponent bit multiplies the value by 2^{128}. Mantissa flips, 72% of all hits, change the value by at most a factor of two.

C. Onboard Earth-Observation Classification

We use land-use/land-cover classification of Sentinel-2 [17] imagery as the workload, with the EuroSAT benchmark [18]: 27,000 RGB chips of 64×64 pixels in ten classes (Annual Crop, Forest, Herbaceous Vegetation, Highway, Industrial, Pasture, Permanent Crop, Residential, River, Sea/Lake). The models are a ResNet-18 [19] and a MobileNetV2 [20], both with batch normalization [21], the two families most often proposed for onboard vision.

III. Related Work

Hardware mitigation. EDAC/ECC, selective or full TMR, and scrubbing are mature and remain the default for flight software memory [7], [10]. ECC adapted to DNN weights has been proposed with substantial area overhead [22], and hybrid redundant accelerators target satellite AI directly [23]. ZEDAC protects model parameters in memory and storage through a rad-tolerant companion [24]. Beam experiments caution that many critical DNN errors originate in logic that ECC does not cover, and that single-bit injection can underestimate real misdetection rates [16]. Our work is complementary: Aegis-v2 assumes a protected golden copy and repairs the unprotected working copy.

Algorithm-based fault tolerance and checksums. ABFT [25] and its convolutional descendants detect errors in computation via checksums; RADAR [14] compares group checksums of weights against golden signatures and zeroes flagged groups; weight reconstruction [26] and WeightSentry [27] defend against targeted bit-flip attacks, which, like radiation, are most damaging in the exponent MSB [9]. We use the simplest member of this family, CRC-32 [28] per block, but only as a confirmer gated by behavior.

Range restriction. Ranger [11], FT-ClipAct [12] and ProAct [29] clamp activations to bounds profiled on clean data. They are cheap and need no storage, but they do not repair weights; under accumulating upsets the clipped network

TABLE I
CAPABILITY COMPARISON. (A) LABEL-FREE BEHAVIORAL MONITOR; (B) DETECTION-GATED RESTORE OF CORRUPTED WEIGHTS; (C) NO LABELS OR GROUND CONTACT NEEDED FOR RECOVERY; (D) EVALUATED UNDER ORBIT-CALIBRATED ACCUMULATING FAULTS.

Approach	(a)	(b)	(c)	(d)
Periodic scrubbing [7]	–	blind	✓	–
Ranger / FT-ClipAct [11], [12]	bounds	–	✓	–
BLINK [13]	BN	–	✓	–
RADAR [14]	–	zeroes group	✓	–
MILR [15]	–	algebraic	✓	–
SHNN [31]	act.	retrain	–	–
RedNet [4]	–	–	✓	flight
Ground re-upload	–	late	–	–
Aegis-v2 (ours)	**bounds+BN**	**CRC-gated**	✓	✓

degrades, which our results confirm for ResNet-18. RedNet combines such tolerance with application awareness and was validated on the Chaohu-1 satellite [4].

Batch-norm based detection. BLINK [13] uses batch-norm statistics as integrity checkpoints to detect weight corruption in situ and re-centers anomalous activations. Our Tier 2 is BLINK-style, and our BLINK baseline implements its re-centering mitigation. The difference is what happens next: Aegis-v2 restores weights rather than compensating for them.

Weight correction and self-healing. MILR [15] recovers layers algebraically from stored input/output relations; CoG-guided correction [30] substitutes profiled values for detected faulty weights; SHNN [31] localizes damage by activation discrepancy and retrains the afflicted layer, which needs labeled data. Replacing corrupted layers from backup copies appears in patent literature [32]. Plasticity methods such as ReDo [33] and continual backpropagation [34] reinitialize dormant units during training; they cannot restore a specific learned function after hardware faults.

Space ML robustness. Lange et al. surveyed the robustness of onboard ML to radiation and found prior work seldom modeled natural hazards for spacecraft models [35]; SENTRY analyzed vision transformers on EuroSAT under soft errors [36]; position papers call for radiation-aware AI runtimes [37]. Fault injectors such as PyTorchFI [38] and Ares [39] established application-level injection as a valid proxy.

Table I summarizes the gap. Every component of Aegis-v2 exists in isolation; to our knowledge the combination, evaluated as a repair *policy* under accumulating, orbit-modulated faults with paired baselines and traffic and energy accounting, has not been reported.

IV. FAULT MODEL AND MISSION SIMULATOR

A. What Is Corrupted

Let the model's stored state be $\theta \in \mathbb{F}_{32}^{N}$, all parameters and batch-norm buffers, occupying $B = 32N$ bits: 358.1 Mbit for ResNet-18 (44.77 MB) and 72.7 Mbit for MobileNetV2 (9.08 MB). We model SEUs in the working copy of θ held in volatile memory. Faults persist until a repair overwrites them. We do not inject into activations, control logic or the golden copy, which we assume resides in radiation-tolerant non-volatile memory (MRAM or rad-hard SPI NOR); transient activation faults are studied separately (Section VII-F).

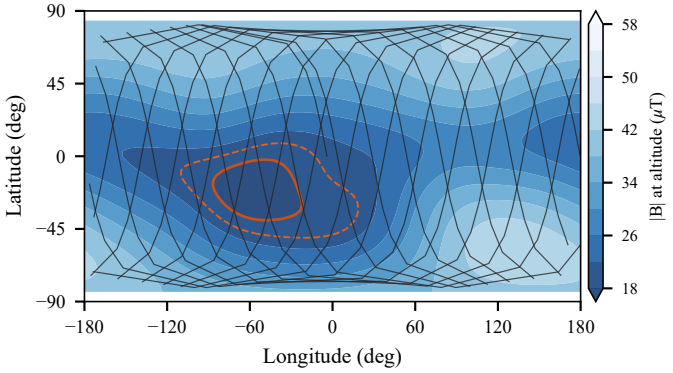


Fig. 2. One day of the Sentinel-2A ground track (SGP4) over IGRF field magnitude at altitude. Solid and dashed contours mark the SAA core (18,000 nT) and edge (20,000 nT) between which the upset rate ramps.

B. Upset Arrival Along a Real Orbit

The satellite follows Sentinel-2A (NORAD 40697, $\approx$ 786 km sun-synchronous) propagated with SGP4 [40] from a CelesTrak two-line element set [41], sampled every minute. At each position $\mathbf{r}(t)$ we evaluate the geomagnetic field magnitude $|B(\mathbf{r})|$ from IGRF-13 [42] and map it to a per-bit upset rate by a linear ramp:

$$\lambda(t) = \lambda_{\mathrm{bg}} + (\lambda_{\mathrm{SAA}} - \lambda_{\mathrm{bg}})\,\mathrm{clip}\left(\frac{B_{\mathrm{e}} - |B(\mathbf{r}(t))|}{B_{\mathrm{e}} - B_{\mathrm{c}}}, 0, 1\right), \quad (1)$$

with edge $B_{\mathrm{e}} = 20{,}000$ nT and core $B_{\mathrm{c}} = 18{,}000$ nT. The number of upsets in a step of length Δ is Poisson with mean $B \int_t^{t+\Delta} \lambda(\tau)\, d\tau$, integrated per minute. Fig. 2 shows one day of the track over the field.

Calibration. We fix the ratio $\lambda_{\mathrm{SAA}}/\lambda_{\mathrm{bg}} = 75.3$ and choose $\lambda_{\mathrm{bg}} = 7.89 \times 10^{-8}$ and $\lambda_{\mathrm{SAA}} = 5.94 \times 10^{-6}$ upsets/bit/day so that the orbit average over the real track (79.9 SAA-core-equivalent minutes per day) reproduces the Alsat-1 mean of 4.04×10^{-7} [10]. About 82% of upsets then occur inside the SAA. As a consistency check, the calibrated rate predicts $358.1 \times 10^{6} \times 4.04 \times 10^{-7} = 145$ flips/day for ResNet-18 and 29.4 for MobileNetV2; the campaigns recorded 147.1 ± 16.6 and 29.1 ± 0.5.

C. Placement and Multiplicity

Each upset event is an SBU with probability 0.9855, otherwise an MBU cluster of up to four adjacent bits. Events are placed uniformly over stored bits: a block is chosen with probability proportional to its bit count, then a tensor, an element and a bit. All 32 bit positions are eligible; restricting injection to the exponent would inflate severity.

D. Paired Replay

A single seeded injector samples the fault stream for the unprotected replica, recording (block, tensor, index, bit) tuples. Every other strategy runs on its own deep copy of the model and *replays the identical records*. Differences between strategies within a seed are therefore paired, and we report paired differences with Student-t intervals.

E. Problem Statement

A repair *strategy* π observes, at each step t, only the unlabeled incoming batch X_t and its own model state, and may rewrite any subset of blocks from the golden copy. Let $A_\pi(t)$ be the top-1 accuracy of its model on a held-out labeled set after step t, $D_\pi(T)$ the bytes it moved and $E_\pi(T)$ the energy it spent over a mission of T steps. We seek strategies that are high in mean accuracy

$$\bar{A}_\pi = \frac{1}{T}\sum_{t=1}^{T} A_\pi(t) \tag{2}$$

and low in $D_\pi(T)$ and $E_\pi(T)$, without labels or ground contact. No single strategy need dominate on all three; we therefore compare strategies on the accuracy–traffic plane and report energy separately. Periodic scrubbing with interval h traces a one-parameter family $\{(\bar{A}_h, D_h)\}$, the frontier that any detection-based policy must beat.

V. The Aegis-v2 Method

Aegis-v2 runs once per monitoring period (one hour in our experiments) on the batch of images the satellite has just acquired. Fig. 3 gives the overview and Algorithm 1 the procedure.

A. Design Evolution: From Aegis-v1 to v2

The design was shaped by what did not work. A first prototype (Aegis-v1) borrowed from plasticity research: it watched for "dead" low-firing neurons in a small multilayer perceptron, modeled faults as zeroed weights, and revived neurons with noise or re-initialization in the spirit of ReDo [33] and continual backpropagation [34]. A prior-art review showed three problems. First, the fault model was wrong: SEUs in float32 weights create huge values, not dead units, so a dead-neuron monitor watches for the opposite symptom. Second, re-initialization re-randomizes capacity but cannot recover a specific learned function without a training signal, and an inference-only payload has neither gradients nor labels. Third, a flat MLP does not represent the batch-normalized CNNs flown in practice. Aegis-v2 keeps the idea of a self-monitoring network but replaces each piece with one grounded in the reliability literature: range and BN-statistic monitors in place of dead-neuron counts (inspired by Ranger [11] and BLINK [13]), checksums in place of heuristics (as in RADAR [14]), and exact restore from a golden copy in place of revival, the operation scrubbing already trusts [7].

B. Blocks

We partition θ into ordered *blocks*, the unit of detection, restoration and accounting: for ResNet-18 the stem, the eight residual blocks `layerX.Y` and the classifier (10 blocks, 38.7 kB to 18.9 MB); for MobileNetV2 the 19 `features.N` stages and the classifier (20 blocks). A single function maps both parameter keys and module names to block identifiers. This matters: an early version used a separate mapping for monitor alarms with a fallback that silently sent every MobileNetV2 alarm to the stem.

C. Tier 1: Activation Envelope

During calibration on $n \geq 100$ clean chips we record, for every convolution and linear layer l, the minimum m_l and maximum M_l of its output. The envelope is widened by a margin $\alpha = 0.5$ of the range:

$$[\ell_l, u_l] = [m_l - \alpha(M_l - m_l),\ M_l + \alpha(M_l - m_l)]. \tag{3}$$

At run time layer l raises an alarm if any output is NaN or $\pm\infty$ or falls outside $[\ell_l, u_l]$. Tier 1 catches exponent-MSB explosions immediately. We stress the choice of envelope: an envelope of mean $\pm k\sigma$ fires on essentially every clean batch because activations are heavy-tailed, which silently degenerates the gate into "restore everything" (Section VIII).

D. Tier 2: Batch-Norm Drift

Each batch-norm layer stores running statistics (μ_c, σ_c^2) per channel c. For the input x of BN layer l on the current batch, with batch statistics $(\hat{\mu}_c, \hat{\sigma}_c)$ and $s_c = \sqrt{\sigma_c^2 + \epsilon}$, $\hat{s}_c = \sqrt{\hat{\sigma}_c^2 + \epsilon}$, the drift score is

$$d_l = \frac{1}{C}\sum_{c=1}^{C}\left(\frac{|\hat{\mu}_c - \mu_c|}{s_c} + \frac{|\hat{s}_c - s_c|}{s_c}\right), \tag{4}$$

set to a sentinel value if any statistic is non-finite. The per-layer threshold is calibrated on clean batches of the run-time batch size as $\tau_l = \max(3.5,\ 2.5\max_{\text{clean}} d_l)$. Tier 2 is sensitive to mid-exponent and large mantissa shifts that keep activations inside the T1 envelope.

E. Tier 3: CRC Confirmation and the Upstream Rule

Behavioral alarms say where to *look*, not what is broken: an exponent flip in one block corrupts every activation downstream, so T1 and T2 typically flag a cascade of blocks. Tier 3 resolves this exactly. At calibration it stores the CRC-32 of each block's serialized parameters; at run time it recomputes the CRC of each candidate and declares a block corrupted iff the checksums differ. Cascade victims pass and are not restored.

T1 hooks convolution outputs and T2 hooks BN *inputs*, so a fault in a BN parameter at the end of block k, e.g. a sign flip on a running variance that produces NaN, first becomes visible at the hooks of block $k{+}1$. Aegis-v2 therefore always adds the block immediately upstream of the earliest candidate to the T3 set. Without this rule a seed in our campaign stayed at 10% accuracy for the rest of the mission because the culprit, `layer3.0.bn2`, was never checksummed.

F. Golden Store and Restore

The golden copy holds one zstd-compressed [43] safetensors blob per block. Trained float32 weights compress poorly (44.77 → 41.58 MB for ResNet-18, 9.08 → 8.44 MB for MobileNetV2), so the store is essentially the model size. Restores are bit-exact, so no batch-norm recalibration [44] is needed; we keep it optional and off, and charge one inference per buffered image when enabled. An INT8 variant (per-output-channel symmetric quantization) shrinks the ResNet-18 store to 9.2 MB at a cost of 0.7 points on 1,000 chips;

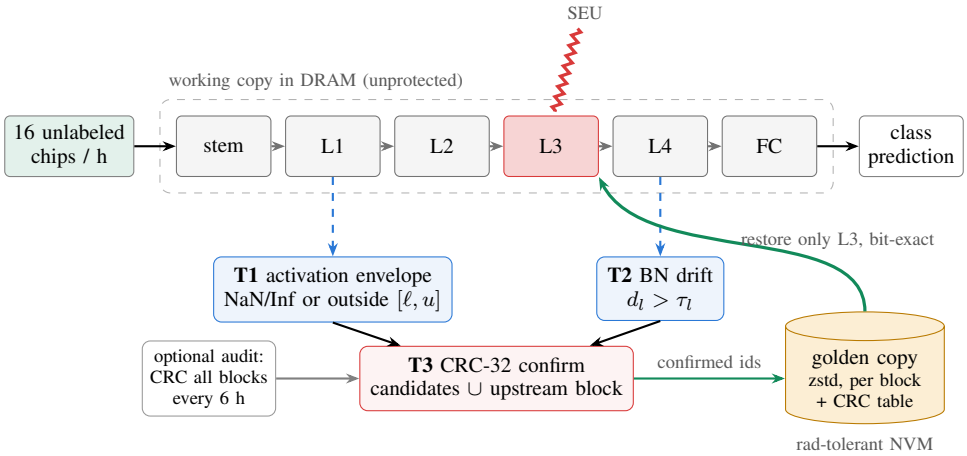


Fig. 3. Aegis-v2 repair loop. Hooks on each incoming batch feed the behavioral tiers T1 and T2, which propose candidate blocks without labels. T3 recomputes CRC-32 over each candidate and the block upstream of the earliest candidate, compares with golden checksums, and only blocks whose stored bits changed are restored from the compressed golden copy. Cascade victims downstream of a corrupted block raise alarms but pass T3 and are left untouched.

Algorithm 1 Aegis-v2 step on an incoming batch

Input: model blocks $b_1, \ldots, b_K$ in forward order; envelopes $[\ell_l, u_l]$; thresholds τ_l; golden CRCs g_k; golden store $\mathcal{G}$
Output: restored block set R

run forward pass on the batch with T1/T2 hooks attached
$C \leftarrow \{\text{block}(l) : \text{T1 alarm at } l \vee d_l > \tau_l\}$
if $C = \emptyset$ **then**
  return $\emptyset$ {healthy: no checksum, no traffic}
end if
$k^* \leftarrow \min\{k : b_k \in C\}$; $C \leftarrow C \cup \{b_{k^*-1}\}$ {upstream rule}
$R \leftarrow \{b_k \in C : \text{CRC32}(b_k) \neq g_k\}$
for all $b_k \in R$ in forward order **do**
  $b_k \leftarrow \text{decompress}(\mathcal{G}[k])$ {bit-exact restore}
end for
return R

its restores are lossy, so restored blocks are re-baselined individually in T3. A pitfall: refreshing all golden checksums while any block may be corrupt bakes the corruption into the reference and makes it permanently invisible; Aegis-v2 only ever re-baselines blocks it has just restored.

G. CRC Audit for Sub-Threshold Faults

Behavioral detection cannot see a fault that keeps activations inside the envelope and statistics inside tolerance. In one seed a single weight changed from 0.106 to 1.7, cost 17 points of accuracy, and raised no alarm. The *Aegis-v2 + CRC audit* variant therefore checksums every block on a fixed period (6 h by default) and restores any mismatch, independently of the behavioral gate. We report it alongside plain Aegis-v2.

H. Combining with Range Restriction

Ranger-style clamping can run under Aegis-v2: monitor hooks are registered before the clamp hooks so that T1 observes raw activations, and the clamps limit damage in the interval before a restore. We evaluate this combination as *Ranger + Aegis-v2*.

I. Cost Model

Every repairing strategy is charged for the bytes it moves and the energy that costs. A restore of block k with compressed size c_k and raw size r_k costs

$$E_{\text{restore}}(k) = 8\, c_k\, e_{\text{flash}} + 8\, r_k\, e_{\text{DRAM,w}}, \tag{5}$$

with $e_{\text{flash}} = 15$ pJ/bit and $e_{\text{DRAM,w}} = 25$ pJ/bit. A T3 checksum of block k costs $8\, r_k\, e_{\text{DRAM,r}}$ with $e_{\text{DRAM,r}} = 20$ pJ/bit. The T1/T2 hooks cost $e_{\text{op}} = 6.9$ pJ per elementwise operation (four passes over each convolution/linear output, three over each BN input), obtained by dividing a 12.5 mJ ResNet-18 inference on a Jetson-class device by its 1.82 G MACs; this treats an elementwise pass as one MAC and is likely pessimistic. Monitoring is charged only on the monitored incoming batch, not on evaluation passes. The ground uplink receives at 1 Mbps during four 480 s contacts per day ($\approx$ 60 MB per pass, as reported for Chaohu-1 [4]) at 500 pJ/bit. Ranger and BLINK move no bytes and their per-inference compute is not charged. All constants are assumptions of order-of-magnitude fidelity (Table II).

VI. Experimental Setup

Data and splits. EuroSAT RGB chips are bicubically upsampled to 224×224 and normalized with EuroSAT channel statistics; logits follow the checkpoint's label order. Calibration (10 chips per class, 100 total) and labeled evaluation (15 per class, 150 total) sets are disjoint. Monitoring operates on a rotating window of 16 unlabeled chips per step; labels are used only to measure accuracy.

Models. ResNet-18 uses a public EuroSAT checkpoint (`cm93/resnet18-eurosat`) with 89.3% accuracy on our evaluation set (92.8% on a 1,000-chip check); its training split is not published, so its clean accuracy may be optimistic,

TABLE II
SIMULATION CONSTANTS. M = MEASURED, C = CALIBRATED, A = ASSUMED.

Quantity	Value	
Orbit-mean upset rate (Alsat-1 [10])	4.04×10^{-7} /bit/day	M
Single-bit fraction	98.55%	M
Background / SAA-core rate	7.89×10^{-8} / 5.94×10^{-6}	C
SAA-core / background ratio	75.3	A
SAA ramp, $\lvert B \rvert$ edge $\rightarrow$ core	20,000 $\rightarrow$ 18,000 nT	A
SAA-core-equivalent time	79.9 min/day	C
Flash read / DRAM write / DRAM read	15 / 25 / 20 pJ/bit	A
Monitor elementwise op	6.9 pJ	A
Inference energy (Jetson-class proxy)	12.5 mJ/image	A
Uplink rate, contacts	1 Mbps, 4×480 s/day	A
RF receive energy	500 pJ/bit	A
T1 margin / T2 floor, factor / audit	0.5 / 3.5, 2.5 / 6 h	–

though relative degradation is unaffected. To remove that caveat on a second architecture we fine-tuned an ImageNet-initialized [45] MobileNetV2 for three epochs (AdamW [46] with a one-cycle schedule [47], batch 64, learning rate 10^{-3}, flip and 90°-rotation augmentation) on 26,035 chips, holding out the 965 chips used by every campaign split; the simulator refuses any split not contained in the held-out set. It reaches 99.3% on the evaluation set.

Mission. Each campaign simulates three days in 72 one-hour steps for three radiation seeds, with the Sentinel-2A track and the rate model of Section IV. Per step the simulator injects the step's upsets, replays them on every replica, runs each strategy's repair logic, and evaluates every replica on the 150 labeled chips.

Strategies. (1) *Unmitigated*. (2) *Ranger clipping*: activations clamped to profiled bounds with a 0.1 margin [11]. (3) *BLINK re-centering*: anomalous BN inputs renormalized to reference statistics (threshold 3.0) [13]. (4) *Periodic scrubbing*: full restore every 12 h, plus a sweep at 1, 3, 6 and 24 h. (5) *Ground uplink*: CRC-confirmed blocks queued and reloaded only during contacts, rate-limited by the link. (6) *Aegis-v2*. (7) *Aegis-v2 + CRC audit* (6 h). (8) *Ranger + Aegis-v2*.

Metrics. The primary metric is *mean accuracy*, the time average of top-1 accuracy over all steps, which reflects the fraction of the mission spent producing correct answers. We also report final accuracy (a single noisy snapshot), repair traffic in MB/day, and modeled energy in mJ/day. Intervals are 95% Student-t over the three seeds ($t_{0.975,2} = 4.30$), so they are wide; paired differences use the same fault stream per seed. The *bandwidth verdict* asks whether any scrubbing interval dominates a strategy (higher accuracy at no more traffic) and how much traffic scrubbing needs to match its accuracy, interpolating linearly between swept intervals.

Implementation. The simulator is written in PyTorch. The final campaigns ran on an NVIDIA T4 (about 200 s per seed for ResNet-18 with 12 model replicas, 225 s for MobileNetV2). The repository includes 51 unit and regression tests, including regression tests for the evaluation pitfalls of Section VIII.

VII. RESULTS

A. Accuracy Under Accumulating Upsets

Table III and Fig. 4 give the main results. On ResNet-18, with 147 upsets per day and 4.9 on the exponent MSB, the unprotected model is at chance for almost the entire mission (mean 14.0%, final 10.0% in every seed). Range clipping and BN re-centering slow the collapse but cannot reverse it, because the corrupted weights stay in memory and accumulate (means 38.5% and 34.0%, ending at 27.1% and 13.6%). Twelve-hour scrubbing returns the model to clean accuracy at each reload but spends most of each interval broken (mean 37.0%). The ground uplink repairs only during contacts (mean 66.2%). Aegis-v2 holds 87.4% against a clean 89.3%; with the CRC audit, 88.9% with a final accuracy equal to clean in every seed.

The gap to plain Aegis-v2 comes from a single event: at hour 49 of seed 1 a sub-threshold weight fault ($0.106 \rightarrow 1.7$) removed 17 points without raising any alarm and persisted to the end of the mission. The audit closes it within six hours at a cost of 49% more traffic and 40% more energy; the paired difference (-1.5 ± 6.6 points) is inside the interval. Combining Ranger with Aegis-v2 gives identical accuracy here and 24% less traffic. Its restore events are as frequent (63 versus 62), so the saving comes from restoring fewer or smaller blocks per event; we have not isolated the mechanism.

MobileNetV2 stores five times fewer bits and is hit five times less often (29 upsets/day). Its ReLU6 activations are bounded by construction, which plausibly explains why range clipping is far more effective on it (98.8%) than on ResNet-18. All three Aegis-v2 variants hold the clean 99.3% in every seed.

B. Paired Comparison

Because every strategy sees the same faults within a seed, Table IV reports paired differences. On ResNet-18 Aegis-v2 beats every non-Aegis-v2 baseline in all three seeds: by +50.5 points over 12-hour scrubbing, +21.2 over the ground uplink, +48.9 over range clipping, +53.4 over BN re-centering and +73.4 over no protection. On MobileNetV2 the differences are smaller and, except against BN re-centering, not significant with three seeds; in seed 2 no fault was damaging enough to separate any strategies, so that seed carries little information.

C. Accuracy per Byte: the Bandwidth Verdict

Scrubbing has one knob, the interval, which trades traffic for accuracy along the curve of Fig. 5. No interval dominates any Aegis-v2 variant on either architecture. Interpolating the frontier, scrubbing needs about 889 MB/day to reach Aegis-v2's 87.4% on ResNet-18, 4.7× the 188 MB/day Aegis-v2 moves; 975 MB/day to match the audit variant (3.5×); and the combination with Ranger reaches the same accuracy at 6.2× less than the matching scrub. On MobileNetV2 the ratio is about 60×, mostly because Aegis-v2 moves almost nothing (3.4 MB/day, nine restore events over all seeds) while the matching 1-hour scrub rewrites the model 24 times a day.

Why is Aegis-v2's ResNet-18 traffic not smaller still? Because the blocks that get hit are the blocks that hold the bits.

TABLE III
MISSION OUTCOMES OVER THREE SIMULATED DAYS (MEAN OF THREE SEEDS, 95% STUDENT-t INTERVALS). REPAIR TRAFFIC AND ENERGY ARE PER SIMULATED DAY; "N.C." = NOT CHARGED (STRATEGY MOVES NO BYTES).

	ResNet-18 (clean 89.3%, 147 flips/day)				**MobileNetV2** (clean 99.3%, 29 flips/day)			
Strategy	Mean acc. (%)	Final acc. (%)	MB/day	mJ/day	Mean acc. (%)	Final acc. (%)	MB/day	mJ/day
Aegis-v2	**87.4**±8.3	83.6±24.9	188.1	144.1±35.8	**99.3**±0.0	99.3±0.0	3.4	125.6±2.5
Aegis-v2 + CRC audit	**88.9**±1.7	89.3±0.0	279.6	202.3±20.3	**99.3**±0.0	99.3±0.0	18.1	136.4±2.5
Ranger + Aegis-v2	**87.4**±8.3	83.6±24.9	143.0	120.5±35.6	**99.3**±0.0	99.3±0.0	1.7	124.7±1.0
Periodic scrubbing (12 h)	37.0±27.0	89.3±0.0	83.2	27.9	87.2±26.1	99.3±0.0	16.9	5.7
Ground uplink reload	66.2±10.2	52.4±91.4	121.5	634.5±167.3	93.3±12.9	99.3±0.0	3.5	141.5±27.7
Ranger clipping	38.5±56.4	27.1±43.1	0	n.c.	98.8±1.4	96.7±6.0	0	n.c.
BLINK re-centering	34.0±42.0	13.6±16.8	0	n.c.	79.5±17.1	60.4±44.9	0	n.c.
Unmitigated	14.0±9.8	10.0±0.0	0	n.c.	76.9±51.7	39.8±128.1	0	n.c.
Scrubbing 1 h	89.3	–	998.0	334.6	99.3	–	202.4	67.9
Scrubbing 3 h	77.6	–	332.7	111.5	97.3	–	67.5	22.6
Scrubbing 6 h	66.4	–	166.3	55.8	96.4	–	33.7	11.3
Scrubbing 24 h	22.0	–	41.6	13.9	85.1	–	8.4	2.8

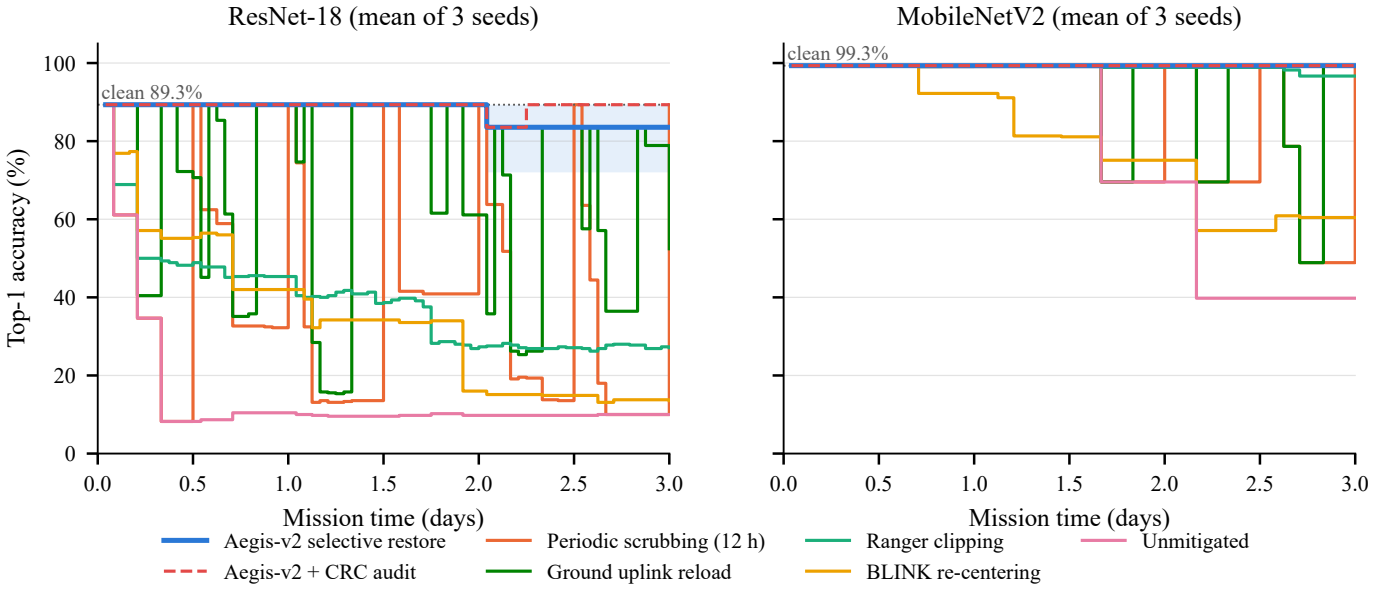


Fig. 4. Accuracy over the mission, averaged over three seeds (shaded: min–max across seeds for Aegis-v2). Unprotected ResNet-18 reaches chance (10%) within eight hours. Scrubbing recovers fully at each 12-hour reload and then decays as new upsets arrive; the ground uplink recovers only during contacts. Aegis-v2 stays at the clean line except where a sub-threshold fault in one seed escapes the behavioral gate (day 2); the CRC audit variant repairs it at the next audit.

TABLE IV
PAIRED DIFFERENCE IN MEAN ACCURACY, AEGIS-V2 MINUS BASELINE (POINTS, 95% INTERVAL), AND SEEDS WHERE AEGIS-V2 WAS HIGHER.

	ResNet-18		MobileNetV2	
Baseline	Δ	wins	Δ	wins
Scrubbing (12 h)	$+50.5 \pm 23.6$	3/3	$+12.1 \pm 26.1$	2/3
Ground uplink	$+21.2 \pm 5.3$	3/3	$+6.0 \pm 12.9$	2/3
Ranger clipping	$+48.9 \pm 48.2$	3/3	$+0.5 \pm 1.4$	2/3
BLINK re-centering	$+53.4 \pm 36.4$	3/3	$+19.8 \pm 17.1$	3/3
Unmitigated	$+73.4 \pm 4.8$	3/3	$+22.4 \pm 51.7$	2/3
Aegis-v2 + audit	-1.5 ± 6.6	0/3	0.0 ± 0.0	0/3

Over the three seeds Aegis-v2 performed 62 restore events covering 162 block restores, of which 92% were the four `layer3`/`layer4` blocks that hold 94% of the parameters (3.7 to 18.9 MB each; restore counts 57, 40, 39 and 13). Layer selectivity saves the most when damage is confined to small blocks; for a model whose parameters concentrate at the end, a restore is a large fraction of a full scrub, and the saving comes mainly from *not* restoring when nothing is wrong.

D. Energy: a Negative Result

Fig. 6 separates the two costs. Under our energy model Aegis-v2 spends 144 mJ/day on ResNet-18 and 126 mJ/day on MobileNetV2, versus 27.9 and 5.7 mJ/day for 12-hour scrubbing: 5.2× and 22× more. Monitoring dominates. The T1/T2 hooks alone cost 46 mJ/day (about 1.9 mJ per 16-image ResNet-18 batch, ten full-model reload equivalents of memory energy over three days), and together with T3 checksum reads monitoring is 56% of Aegis-v2's energy on ResNet-18 and 98% on MobileNetV2, whose many narrow layers make elementwise hooks relatively expensive. Our first prototype

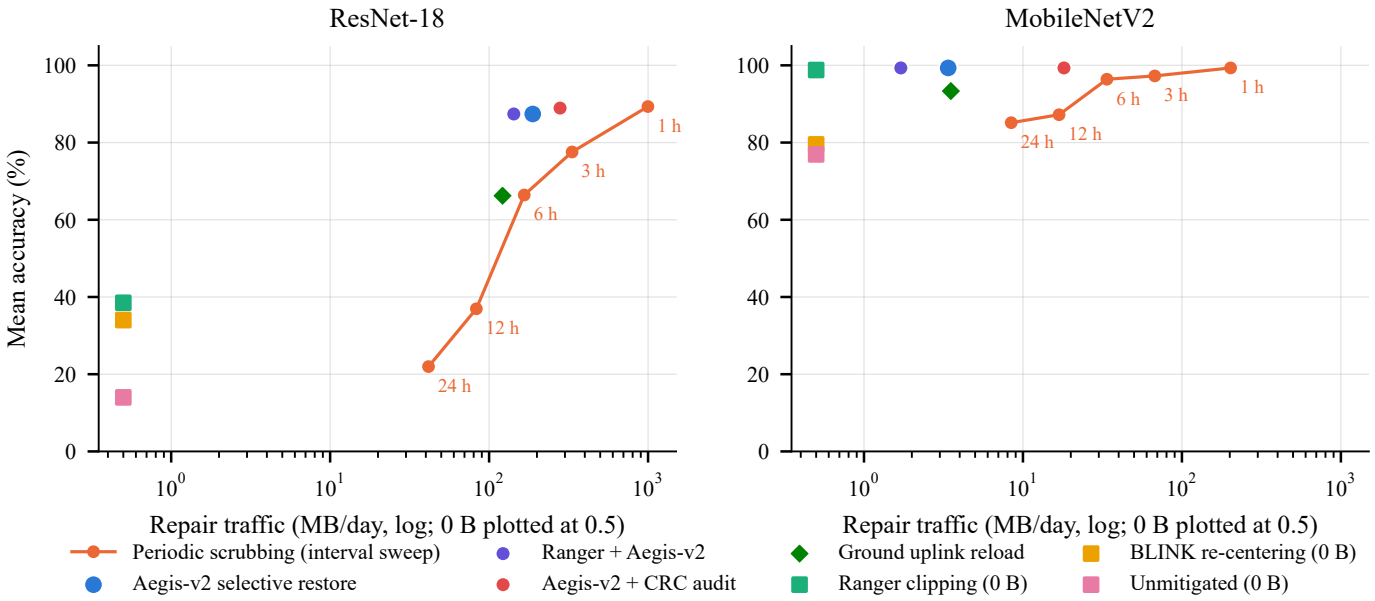


Fig. 5. Mean accuracy versus repair traffic. The orange curve sweeps the scrubbing interval from 24 h to 1 h; strategies that move no bytes are drawn at 0.5 MB/day. All Aegis-v2 variants lie above and to the left of the scrubbing frontier on both architectures.

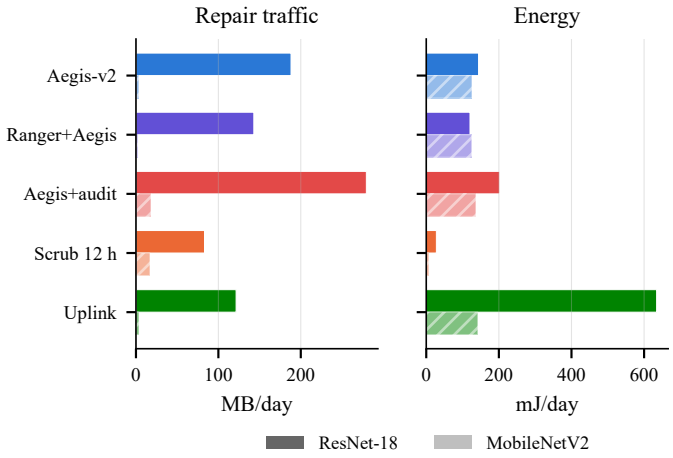


Fig. 6. Repair traffic and modeled energy per day. Solid: ResNet-18; hatched: MobileNetV2. Aegis-v2's energy is dominated by always-on monitoring, not by restores.

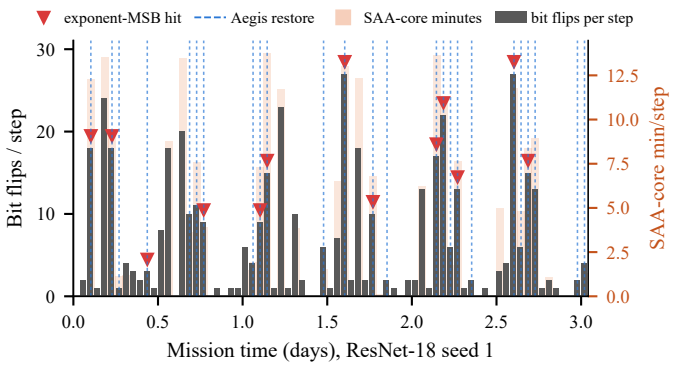


Fig. 7. Upsets per one-hour step for ResNet-18 seed 1 against SAA-core minutes in the step. Upsets cluster in the SAA passes; exponent-MSB hits (triangles) arrive about five times a day; dashed lines mark steps where Aegis-v2 restored at least one block.

claimed a 98.9% energy saving because it did not charge the monitor; that claim was wrong. With the monitor charged, Aegis-v2 is an energy *cost*, and our honest claim is about bandwidth and availability. Two caveats cut in its favor: the 6.9 pJ per elementwise operation is likely pessimistic for fused hardware reductions, and monitoring could run on every n-th batch, trading detection latency for energy linearly. The ground uplink is the most expensive option by far (635 mJ/day on ResNet-18) because radio reception costs an order of magnitude more per bit than memory.

E. Where Faults Land and What Gets Restored

Fig. 7 shows the arrival process for one seed. Upsets bunch into the consecutive daily SAA passes, and quiet hours alternate with bursts of 20–30 flips. Of the 1,324 ResNet-18 upsets over three seeds, 72.5% landed in mantissa bits, 24.6% in exponent bits and 2.9% in sign bits, close to the 23/32, 8/32 and 1/32 expected from uniform placement. Restores coincide with exponent-MSB hits and with other damaging hits (sign flips on BN variances, high exponent bits), and Aegis-v2 is silent through the many steps whose upsets touch only mantissas.

F. Transient Faults Do Not Trigger Restores

A behavioral monitor alone cannot distinguish a transient glitch in an activation buffer from corrupted weights. We injected exponent-MSB flips into activations, not weights, in 40 single-batch trials on ResNet-18. Tier 1 or 2 raised an alarm in 27.5% of trials; Tier 3 found no checksum mismatch in any, so the spurious-restore rate was 0% and no bytes were moved. Mean accuracy over the trials changed from 89.7% clean to 88.6% with the transient fault (the glitch affects one batch

only), and to 89.7% with range clipping. This is the second purpose of the CRC gate: beyond localizing cascades, it makes repair decisions robust to faults that need no repair.

VIII. Lessons: Pitfalls That Invalidated Results

Each of the following silently produced a plausible but wrong result during this work. We list them because each is easy to reproduce elsewhere; the fixes are covered by regression tests in the repository.

1) **Synthetic inputs.** Evaluating on Gaussian-noise tensors gave accuracy numbers with no meaning; our first headline ("98.9% energy savings") came from random images, an always-firing gate and an uncharged monitor.
2) **Input contract.** The checkpoint expects 224 px bicubic-upsampled chips with dataset statistics and its own label order. 64 px input gives $\approx 18\%$; alphabetical labels give chance.
3) **Envelope statistics.** A mean$\pm k\sigma$ activation envelope fires on every clean batch because activations are heavy-tailed; the "detection-gated" method then restores on every step. Bounds must be observed extrema plus margin over ≥ 100 calibration chips.
4) **Calibration leakage.** Calibration and evaluation chips must be disjoint; and a model trained by the authors must exclude both (enforced for MobileNetV2).
5) **Poisoned references.** Re-computing golden checksums while a block may be corrupt makes that corruption undetectable forever.
6) **Hook placement.** T1/T2 observe a BN-parameter fault one block downstream; without the upstream rule the culprit is never checksummed.
7) **Name mapping.** Two code paths mapping layer names to blocks diverged for MobileNetV2 and sent every alarm to the stem.
8) **Fault field.** Restricting injection to exponent bits raises the rate of exponent-MSB hits fourfold relative to uniform placement; any such choice must be disclosed.
9) **Uncharged costs.** Omitting monitor compute, CRC reads or recalibration inferences turns an energy cost into an apparent saving.

IX. Discussion

Why gating works. Most upsets are harmless: 72% hit mantissas and change a weight by less than a factor of two. A blind scrubber pays for all of them; an in-place mitigator pays for none of them but cannot undo the rare catastrophic one. Aegis-v2 pays only when behavior *and* checksums agree that something is broken, which is what lets it sit above the scrubbing frontier.

When it does not. The behavioral gate has a blind spot, faults large enough to matter but too small to move activations past calibrated bounds. The audit closes it at a cost that scales with model size and audit frequency; a mission could schedule audits after SAA exits, when new faults are most likely, instead of on a fixed clock. Gating also cannot help when monitoring itself dominates energy (MobileNetV2), and it cannot repair the golden copy, whose integrity rests on rad-tolerant storage and its own ECC.

Deployment. The golden copy fits comfortably in MRAM or rad-hard NOR for models of this size, or in 9.2 MB with INT8 at a small accuracy cost. T3 needs only a CRC engine and a table of 32-bit references; T1/T2 need reductions over activations that many accelerators can fuse into the layers that produce them. The method is complementary to ECC: with ECC on the working memory, Aegis-v2 would act on the residue of uncorrectable and logic-induced errors.

X. Threats to Validity

Radiation model. Rates are transferred from Alsat-1 (686 km, 2002-era SRAM) to Sentinel-2A at 786 km and to modern memory, whose sensitivity can differ by orders of magnitude. The SAA-to-background ratio is assumed, the IGRF ramp is a proxy for a trapped-proton flux model such as AP8/AE9 combined with device cross-sections, solar-cycle variation is not modeled, and faults are injected only into parameters, not into control logic or activation buffers, which beam studies show can dominate [16]. **Statistics.** Three seeds and 150 evaluation chips give wide intervals; one chip is 0.67 points. The MobileNetV2 paired differences are mostly not significant. **Energy.** All energy constants are analytic proxies, not measurements on flight hardware, and Ranger/BLINK compute is not charged, which flatters those baselines. **Models.** The ResNet-18 checkpoint's training split is unknown; MobileNetV2 has a verified held-out split. **Mission length.** Three days captures several SAA cycles and dozens of exponent-MSB hits, but not long-term effects such as total ionizing dose. **Literature.** Several closely related works are 2026 preprints whose results we cite as reported.

XI. Future Work

Natural extensions are (i) orbit-aware scheduling of audits and monitoring duty cycle around SAA passes, which directly targets both the blind spot and the energy cost; (ii) a physics-based environment (AP8/AE9 with device cross-sections via SPENVIS or CREME96); (iii) measured energy and latency on a Jetson- or NPU-class board; (iv) activation and logic faults alongside weight faults; (v) vision transformers and larger EO tasks such as segmentation; and (vi) class-level analysis of which semantic confusions (e.g. River vs. Highway) radiation induces, for which our simulator already records per-class predictions.

XII. Conclusion

Onboard classifiers in LEO accumulate silent, sometimes catastrophic weight corruption. We showed that combining cheap behavioral signals with exact checksums lets a satellite repair only what is broken, when it is broken: on an orbit-calibrated, paired simulation of a Sentinel-2A mission, Aegis-v2 kept a ResNet-18 within two points of clean accuracy where 12-hour scrubbing, range clipping, re-centering and ground reload lost 23 to 55 points, and it did so with several times less repair traffic than the scrubbing interval that would match it. The same analysis also showed what it does not do: with

honest accounting, always-on monitoring costs more energy than a slow scrub. We hope the simulator, the pitfalls and the results help the next design start from realistic faults and complete budgets.

ACKNOWLEDGMENT

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APPENDIX
REPRODUCIBILITY

All results in this paper come from the campaigns `results/main_final` and `results/main_final_mobilenet` in the repository. To regenerate:

```
python experiments/run_campaign.py --days 3 \
  --seeds 0 1 2 --eval-per-class 15 \
  --scrub-sweep-hours 1 3 6 24 --name main_final
python experiments/train_mobilenet.py --epochs 3
python experiments/run_campaign.py ... \
  --arch mobilenet_v2 --name main_final_mobilenet
python experiments/write_report.py --name main_final
python researchpaper/make_paper_figures.py
```

Every seed file records the fault field, orbit model, golden-store format and recalibration flag with its numbers.

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